

and the crawls. See Grimshaw (1990, pp. 161–163) for a readable derivation of this wonderful formula, discovered by Mary Cartwright (1952). See also Stoker (1950) for more about relaxation oscillations.

One last remark: We have seen that a relaxation oscillation has two time scales that operate *sequentially*—a slow buildup is followed by a fast discharge. In the next section we will encounter problems where two time scales operate *concurrently*, and that makes the problems a bit more subtle.

7.6 Weakly Nonlinear Oscillators

This section deals with equations of the form

$$\ddot{x} + x + \varepsilon h(x, \dot{x}) = 0 \quad (1)$$

where $0 \leq \varepsilon \ll 1$ and $h(x, \dot{x})$ is an arbitrary smooth function. Such equations represent small perturbations of the linear oscillator $\ddot{x} + x = 0$ and are therefore called **weakly nonlinear oscillators**. Two fundamental examples are the van der Pol equation

$$\ddot{x} + x + \varepsilon(x^2 - 1)\dot{x} = 0, \quad (2)$$

(now in the limit of small nonlinearity), and the **Duffing equation**

$$\ddot{x} + x + \varepsilon x^3 = 0. \quad (3)$$

To illustrate the kinds of phenomena that can arise, Figure 7.6.1 shows a computer-generated solution of the van der Pol equation in the $(x, \dot{x})$ phase plane, for $\varepsilon = 0.1$ and an initial condition close to the origin. The trajectory is a slowly winding spiral; it takes many cycles for the amplitude to grow substantially. Eventually the trajectory asymptotes to an approximately circular limit cycle whose radius is close to 2.

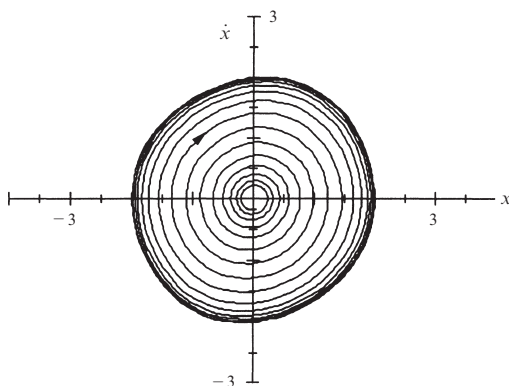


Figure 7.6.1

We'd like to be able to predict the shape, period, and radius of this limit cycle. Our analysis will exploit the fact that the oscillator is “close to” a simple harmonic oscillator, which we understand completely.

Regular Perturbation Theory and Its Failure

As a first approach, we seek solutions of (1) in the form of a power series in ε . Thus if $x(t, \varepsilon)$ is a solution, we expand it as

$$x(t, \varepsilon) = x_0(t) + \varepsilon x_1(t) + \varepsilon^2 x_2(t) + \dots, \quad (4)$$

where the unknown functions $x_k(t)$ are to be determined from the governing equation and the initial conditions. The hope is that all the important information is captured by the first few terms—ideally, the first *two*—and that the higher-order terms represent only tiny corrections. This technique is called **regular perturbation theory**. It works well on certain classes of problems (for instance, Exercise 7.3.9), but as we'll see, it runs into trouble here.

To expose the source of the difficulties, we begin with a practice problem that can be solved exactly. Consider the weakly damped linear oscillator

$$\ddot{x} + 2\varepsilon\dot{x} + x = 0, \quad (5)$$

with initial conditions

$$x(0) = 0, \quad \dot{x}(0) = 1. \quad (6)$$

Using the techniques of Chapter 5, we find the exact solution

$$x(t, \varepsilon) = (1 - \varepsilon^2)^{-1/2} e^{-\varepsilon t} \sin[(1 - \varepsilon^2)^{1/2} t]. \quad (7)$$

Now let's solve the same problem using perturbation theory. Substitution of (4) into (5) yields

$$\frac{d^2}{dt^2}(x_0 + \varepsilon x_1 + \dots) + 2\varepsilon \frac{d}{dt}(x_0 + \varepsilon x_1 + \dots) + (x_0 + \varepsilon x_1 + \dots) = 0. \quad (8)$$

If we group the terms according to powers of ε , we get

$$[\ddot{x}_0 + x_0] + \varepsilon[\ddot{x}_1 + 2\dot{x}_0 + x_1] + O(\varepsilon^2) = 0. \quad (9)$$

Since (9) is supposed to hold for *all* sufficiently small ε , the coefficients of each power of ε must vanish separately. Thus we find

$$O(1): \quad \ddot{x}_0 + x_0 = 0 \quad (10)$$

$$O(\varepsilon): \quad \ddot{x}_1 + 2\dot{x}_0 + x_1 = 0. \quad (11)$$

(We're ignoring the $O(\varepsilon^2)$ and higher equations, in the optimistic spirit mentioned earlier.)

The appropriate initial conditions for these equations come from (6). At $t = 0$, (4) implies that $0 = x_0(0) + \varepsilon x_1(0) + \dots$; this holds for all ε , so

$$x_0(0) = 0, \quad x_1(0) = 0. \quad (12)$$

By applying a similar argument to $\dot{x}(0)$ we obtain

$$\dot{x}_0(0) = 1, \quad \dot{x}_1(0) = 0. \quad (13)$$

Now we solve the initial-value problems one by one; they fall like dominoes. The solution of (10), subject to the initial conditions $x_0(0) = 0$, $\dot{x}_0(0) = 1$, is

$$x_0(t) = \sin t. \quad (14)$$

Plugging this solution into (11) gives

$$\ddot{x}_1 + x_1 = -2 \cos t. \quad (15)$$

Here's the first sign of trouble: the right-hand side of (15) is a **resonant** forcing. The solution of (15) subject to $x_1(0) = 0$, $\dot{x}_1(0) = 0$ is

$$x_1(t) = -t \sin t, \quad (16)$$

which is a **secular** term, i.e., a term that *grows* without bound as $t \rightarrow \infty$.

In summary, the solution of (5), (6) according to perturbation theory is

$$x(t, \varepsilon) = \sin t - \varepsilon t \sin t + O(\varepsilon^2). \quad (17)$$

How does this compare with the exact solution (7)? In Exercise 7.6.1, you are asked to show that the two formulas agree in the following sense: If (7) is expanded as power series in ε , the first two terms are given by (17). In fact, (17) is the beginning of a *convergent* series expansion for the true solution. For any fixed t , (17) provides a good approximation as long as ε is small enough—specifically, we need $\varepsilon t \ll 1$ so that the correction term (which is actually $O(\varepsilon^2 t^2)$) is negligible.

But normally we are interested in the behavior for *fixed* ε , not fixed t . In that case we can only expect the perturbation approximation to work for times $t \ll O(1/\varepsilon)$. To illustrate this limitation, Figure 7.6.2 plots the exact solution (7) and the perturbation series (17) for $\varepsilon = 0.1$. As expected, the perturbation series works reasonably well if $t \ll \frac{1}{\varepsilon} = 10$, but it breaks down after that.

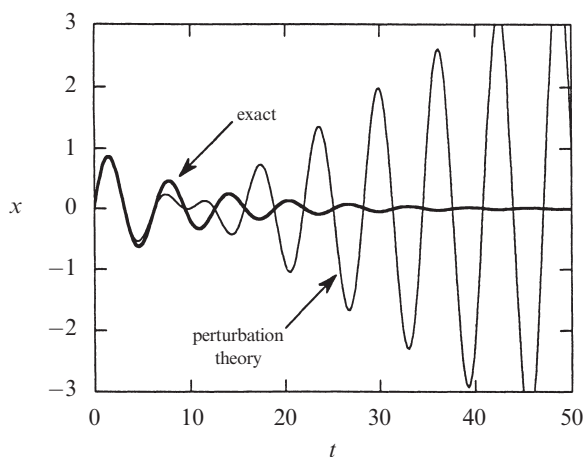


Figure 7.6.2

In many situations we'd like our approximation to capture the true solution's qualitative behavior for all t , or at least for large t . By this criterion, (17) is a failure, as Figure 7.6.2 makes obvious. There are two major problems:

1. The true solution (7) exhibits **two time scales**: a *fast time* $t \sim O(1)$ for the sinusoidal oscillations and a *slow time* $t \sim 1/\varepsilon$ over which the amplitude decays. Equation (17) completely misrepresents the slow time scale behavior. In particular, because of the secular term $t \sin t$, (17) falsely suggests that the solution grows with time whereas we know from (7) that the amplitude $A = (1 - \varepsilon^2)^{-1/2} e^{-\varepsilon t}$ decays exponentially.

The discrepancy occurs because $e^{-\varepsilon t} = 1 - \varepsilon t + O(\varepsilon^2 t^2)$, so to this order in ε , it appears (incorrectly) that the amplitude increases with t . To get the correct result, we'd need to calculate an *infinite* number of terms in the series. That's worthless; we want series approximations that work well with just one or two terms.

2. The frequency of the oscillations in (7) is $\omega = (1 - \varepsilon^2)^{1/2} \approx 1 - \frac{1}{2}\varepsilon^2$, which is shifted slightly from the frequency $\omega = 1$ of (17). After a *very* long time $t \sim O(1/\varepsilon^2)$, this frequency error will have a significant cumulative effect. Note that this is a third, *super-slow* time scale!

Two-Timing

The elementary example above reveals a more general truth: There are going to be (at least) two time scales in weakly nonlinear oscillators. We've already met this phenomenon in Figure 7.6.1, where the amplitude of the spiral grew very slowly compared to the cycle time. An analytical method called **two-timing** builds in the fact of two time scales from the start, and produces better approximations than

regular perturbation theory. In fact, more than two times can be used, but we'll stick to the simplest case.

To apply two-timing to (1), let $\tau = t$ denote the fast $O(1)$ time, and let $T = \varepsilon t$ denote the slow time. We'll treat these two times as if they were *independent* variables. In particular, functions of the slow time T will be regarded as *constants* on the fast time scale τ . It's hard to justify this idea rigorously, but it works! (Here's an analogy: it's like saying that your height is constant on the time scale of a day. Of course, over many months or years your height can change dramatically, especially if you're an infant or a pubescent teenager, but over one day your height stays constant, to a good approximation.)

Now we turn to the mechanics of the method. We expand the solution of (1) as a series

$$x(t, \varepsilon) = x_0(\tau, T) + \varepsilon x_1(\tau, T) + O(\varepsilon^2). \quad (18)$$

The time derivatives in (1) are transformed using the chain rule:

$$\dot{x} = \frac{dx}{dt} = \frac{\partial x}{\partial \tau} + \frac{\partial x}{\partial T} \frac{\partial T}{\partial t} = \frac{\partial x}{\partial \tau} + \varepsilon \frac{\partial x}{\partial T}. \quad (19)$$

A subscript notation for differentiation is more compact; thus we write (19) as

$$\dot{x} = \partial_\tau x + \varepsilon \partial_T x. \quad (20)$$

After substituting (18) into (20) and collecting powers of ε , we find

$$\dot{x} = \partial_\tau x_0 + \varepsilon (\partial_\tau x_0 + \partial_\tau x_1) + O(\varepsilon^2). \quad (21)$$

Similarly,

$$\ddot{x} = \partial_{\tau\tau} x_0 + \varepsilon (\partial_{\tau\tau} x_1 + 2\partial_{T\tau} x_0) + O(\varepsilon^2). \quad (22)$$

To illustrate the method, let's apply it to our earlier test problem.

EXAMPLE 7.6.1:

Use two-timing to approximate the solution to the damped linear oscillator $\ddot{x} + 2\varepsilon\dot{x} + x = 0$, with initial conditions $x(0) = 0$, $\dot{x}(0) = 1$.

Solution: After substituting (21) and (22) for $\dot{x}$ and $\ddot{x}$, we get

$$\partial_{\tau\tau} x_0 + \varepsilon (\partial_{\tau\tau} x_1 + 2\partial_{T\tau} x_0) + 2\varepsilon \partial_\tau x_0 + x_0 + \varepsilon x_1 + O(\varepsilon^2) = 0. \quad (23)$$

Collecting powers of ε yields a pair of differential equations:

$$O(1): \quad \partial_{\tau\tau} x_0 + x_0 = 0 \quad (24)$$

$$O(\varepsilon): \quad \partial_{\tau\tau} x_1 + 2\partial_{T\tau} x_0 + 2\partial_\tau x_0 + x_1 = 0. \quad (25)$$

Equation (24) is just a simple harmonic oscillator. Its general solution is

$$x_0 = A \sin \tau + B \cos \tau, \quad (26)$$

but now comes the interesting part: *The “constants” A and B are actually functions of the slow time T .* Here we are invoking the above-mentioned ideas that τ and T should be regarded as independent variables, with functions of T behaving like constants on the fast time scale τ .

To determine $A(T)$ and $B(T)$, we need to go to the next order of ε . Substituting (26) into (25) gives

$$\begin{aligned} \partial_{\tau\tau} x_1 + x_1 &= -2(\partial_{T\tau} x_0 + \partial_{\tau} x_0) \\ &= -2(A' + A)\cos \tau + 2(B' + B)\sin \tau \end{aligned} \quad (27)$$

where the prime denotes differentiation with respect to T .

Now we face the same predicament that ruined us after (15). As in that case, the right-hand side of (27) is a resonant forcing that will produce *secular terms* like $\tau \sin \tau$ and $\tau \cos \tau$ in the solution for x_1 . These terms would lead to a convergent but useless series expansion for x . Since we want an approximation free from secular terms, *we set the coefficients of the resonant terms to zero*—this maneuver is characteristic of all two-timing calculations. Here it yields

$$A' + A = 0 \quad (28)$$

$$B' + B = 0. \quad (29)$$

The solutions of (28) and (29) are

$$A(T) = A(0)e^{-T}$$

$$B(T) = B(0)e^{-T}.$$

The last step is to find the initial values $A(0)$ and $B(0)$. They are determined by (18), (26), and the given initial conditions $x(0) = 0$, $\dot{x}(0) = 1$, as follows. Equation (18) gives $0 = x(0) = x_0(0,0) + \varepsilon x_1(0,0) + O(\varepsilon^2)$. To satisfy this equation for *all* sufficiently small ε , we must have

$$x_0(0,0) = 0 \quad (30)$$

and $x_1(0,0) = 0$. Similarly,

$$1 = \dot{x}(0) = \partial_{\tau} x_0(0,0) + \varepsilon (\partial_T x_0(0,0) + \partial_{\tau} x_1(0,0)) + O(\varepsilon^2)$$

so

$$\partial_{\tau} x_0(0,0) = 1 \quad (31)$$

and $\partial_T x_0(0,0) + \partial_\tau x_1(0,0) = 0$. Combining (26) and (30) we find $B(0) = 0$; hence $B(T) \equiv 0$. Similarly, (26) and (31) imply $A(0) = 1$, so $A(T) = e^{-T}$. Thus (26) becomes

$$x_0(\tau, T) = e^{-T} \sin \tau. \quad (32)$$

Hence

$$\begin{aligned} x &= e^{-T} \sin \tau + O(\varepsilon) \\ &= e^{-\varepsilon t} \sin t + O(\varepsilon) \end{aligned} \quad (33)$$

is the approximate solution predicted by two-timing. ■

Figure 7.6.3 compares the two-timing solution (33) to the exact solution (7) for $\varepsilon = 0.1$. The two curves are almost indistinguishable, even though ε is not terribly small. This is a characteristic feature of the method—it often works better than it has any right to.

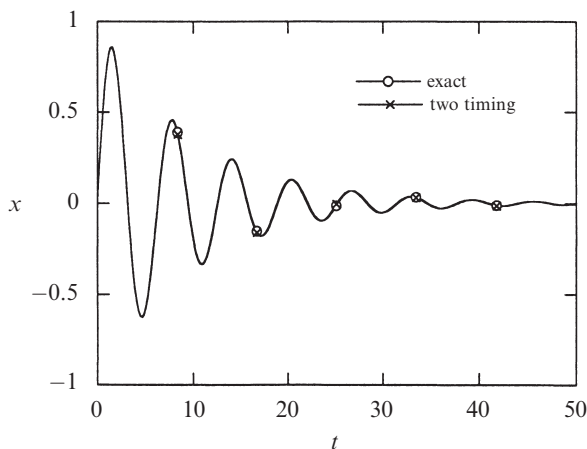


Figure 7.6.3

If we wanted to go further with Example 7.6.1, we could either solve for x_1 and higher-order corrections, or introduce a super-slow time $\mathfrak{T} = \varepsilon^2 t$ to investigate the long-term phase shift caused by the $O(\varepsilon^2)$ error in frequency. But Figure 7.6.3 shows that we already have a good approximation.

OK, enough practice problems! Now that we have calibrated the method, let's unleash it on a genuine nonlinear problem.

EXAMPLE 7.6.2:

Use two-timing to show that the van der Pol oscillator (2) has a stable limit cycle that is nearly circular, with a radius $= 2 + O(\varepsilon)$ and a frequency $\omega = 1 + O(\varepsilon^2)$.

Solution: The equation is $\ddot{x} + x + \varepsilon(x^2 - 1)\dot{x} = 0$. Using (21) and (22) and collecting powers of ε , we find the following equations:

$$O(1): \quad \partial_{\tau\tau}x_0 + x_0 = 0 \quad (34)$$

$$O(\varepsilon): \quad \partial_{\tau\tau}x_1 + x_1 = -2\partial_{\tau}x_0 - (x_0^2 - 1)\partial_{\tau}x_0. \quad (35)$$

As always, the $O(1)$ equation is a simple harmonic oscillator. Its general solution can be written as (26), or alternatively, as

$$x_0 = r(T)\cos(\tau + \phi(T)) \quad (36)$$

where $r(T)$ and $\phi(T)$ are the *slowly-varying amplitude and phase* of x_0 .

To find equations governing r and ϕ , we insert (36) into (35). This yields

$$\begin{aligned} \partial_{\tau\tau}x_1 + x_1 = & -2(r'\sin(\tau + \phi) + r\phi'\cos(\tau + \phi)) \\ & - r\sin(\tau + \phi)[r^2\cos^2(\tau + \phi) - 1]. \end{aligned} \quad (37)$$

As before, we need to avoid resonant terms on the right-hand side. These are terms proportional to $\cos(\tau + \phi)$ and $\sin(\tau + \phi)$. Some terms of this form already appear explicitly in (37). But—and this is the important point—there is also a resonant term lurking in $\sin(\tau + \phi)\cos^2(\tau + \phi)$, because of the trigonometric identity

$$\sin(\tau + \phi)\cos^2(\tau + \phi) = \frac{1}{4}[\sin(\tau + \phi) + \sin 3(\tau + \phi)]. \quad (38)$$

(Exercise 7.6.10 reminds you how to derive such identities, but usually we won't need them—shortcuts are available, as we'll see.) After substituting (38) into (37), we get

$$\begin{aligned} \partial_{\tau\tau}x_1 + x_1 = & \left[-2r' + r - \frac{1}{4}r^3\right]\sin(\tau + \phi) \\ & + [-2r\phi']\cos(\tau + \phi) - \frac{1}{4}r^3\sin 3(\tau + \phi). \end{aligned} \quad (39)$$

To avoid secular terms, we require

$$-2r' + r - \frac{1}{4}r^3 = 0 \quad (40)$$

$$-2r\phi' = 0. \quad (41)$$

First consider (40). It may be rewritten as a vector field

$$r' = \frac{1}{8}r(4 - r^2) \quad (42)$$

on the half-line $r \geq 0$. Following the methods of Chapter 2 or Example 7.1.1, we see that $r^* = 0$ is an unstable fixed point and $r^* = 2$ is a stable fixed point. Hence

$r(T) \rightarrow 2$ as $T \rightarrow \infty$. Secondly, (41) implies $\phi' = 0$, so $\phi(T) = \phi_0$ for some constant ϕ_0 . Hence $x_0(\tau, T) \rightarrow 2\cos(\tau + \phi_0)$ and therefore

$$x(t) \rightarrow 2\cos(t + \phi_0) + O(\varepsilon) \quad (43)$$

as $t \rightarrow \infty$. Thus $x(t)$ approaches a stable limit cycle of radius $= 2 + O(\varepsilon)$.

To find the frequency implied by (43), let $\theta = t + \phi(T)$ denote the argument of the cosine. Then the angular frequency ω is given by

$$\omega = \frac{d\theta}{dt} = 1 + \frac{d\phi}{dT} \frac{dT}{dt} = 1 + \varepsilon\phi' = 1, \quad (44)$$

through first order in ε . Hence $\omega = 1 + O(\varepsilon^2)$; if we want an explicit formula for this $O(\varepsilon^2)$ correction term, we'd need to introduce a super-slow time $\mathfrak{T} = \varepsilon^2 t$, or we could use the Poincaré-Lindstedt method, as discussed in the exercises. ■

Averaged Equations

The same steps occur again and again in problems about weakly nonlinear oscillators. We can save time by deriving some general formulas.

Consider the equation for a general weakly nonlinear oscillator:

$$\ddot{x} + x + \varepsilon h(x, \dot{x}) = 0. \quad (45)$$

The usual two-timing substitutions give

$$O(1): \quad \partial_{\tau\tau} x_0 + x_0 = 0 \quad (46)$$

$$O(\varepsilon): \quad \partial_{\tau\tau} x_1 + x_1 = -2\partial_{\tau T} x_0 - h \quad (47)$$

where now $h = h(x_0, \partial_{\tau} x_0)$. As in Example 7.6.2, the solution of the $O(1)$ equation is

$$x_0 = r(T)\cos(\tau + \phi(T)). \quad (48)$$

Our goal is to derive differential equations for r' and ϕ' , analogous to (40) and (41). We'll find these equations by insisting, as usual, that there be no terms proportional to $\cos(\tau + \phi)$ and $\sin(\tau + \phi)$ on the right-hand side of (47). Substituting (48) into (47), we see that this right-hand side is

$$2[r' \sin(\tau + \phi) + r\phi' \cos(\tau + \phi)] - h \quad (49)$$

where now $h = h(r\cos(\tau + \phi), -r\sin(\tau + \phi))$.

To extract the terms in h proportional to $\cos(\tau + \phi)$ and $\sin(\tau + \phi)$, we borrow some ideas from Fourier analysis. (If you're unfamiliar with Fourier analysis, don't worry—we'll derive all that we need in Exercise 7.6.12.) Notice that h is a 2π -periodic function of $\tau + \phi$. Let

$$\theta = \tau + \phi.$$

Fourier analysis tells us that $h(\theta)$ can be written as a **Fourier series**

$$h(\theta) = \sum_{k=0}^{\infty} a_k \cos k\theta + \sum_{k=1}^{\infty} b_k \sin k\theta \quad (50)$$

where the **Fourier coefficients** are given by

$$\begin{aligned} a_0 &= \frac{1}{2\pi} \int_0^{2\pi} h(\theta) d\theta \\ a_k &= \frac{1}{\pi} \int_0^{2\pi} h(\theta) \cos k\theta d\theta, \quad k \geq 1 \\ b_k &= \frac{1}{\pi} \int_0^{2\pi} h(\theta) \sin k\theta d\theta, \quad k \geq 1. \end{aligned} \quad (51)$$

Hence (49) becomes

$$2[r' \sin \theta + r\phi' \cos \theta] - \sum_{k=0}^{\infty} a_k \cos k\theta - \sum_{k=1}^{\infty} b_k \sin k\theta. \quad (52)$$

The only resonant terms in (52) are $[2r' - b_1] \sin \theta$ and $[2r\phi' - a_1] \cos \theta$. Therefore, to avoid secular terms we need $r' = b_1/2$ and $r\phi' = a_1/2$. Using the expressions in (51) for a_1 and b_1 , we obtain

$$\begin{aligned} r' &= \frac{1}{2\pi} \int_0^{2\pi} h(\theta) \sin \theta d\theta \equiv \langle h \sin \theta \rangle \\ r\phi' &= \frac{1}{2\pi} \int_0^{2\pi} h(\theta) \cos \theta d\theta \equiv \langle h \cos \theta \rangle \end{aligned} \quad (53)$$

where the angled brackets $\langle \cdot \rangle$ denote an average over one cycle of θ .

The equations in (53) are called the **averaged** or **slow-time equations**. To use them, we write out $h = h(r \cos(\tau + \phi), -r \sin(\tau + \phi)) = h(r \cos \theta, -r \sin \theta)$ explicitly, and then compute the relevant averages over the fast variable θ , treating the slow variable r as constant. Here are some averages that appear often:

$$\begin{aligned} \langle \cos \rangle &= \langle \sin \rangle = 0, \quad \langle \sin \cos \rangle = 0, \quad \langle \cos^3 \rangle = \langle \sin^3 \rangle = 0, \quad \langle \cos^{2n+1} \rangle = \langle \sin^{2n+1} \rangle = 0, \\ \langle \cos^2 \rangle &= \langle \sin^2 \rangle = \frac{1}{2}, \quad \langle \cos^4 \rangle = \langle \sin^4 \rangle = \frac{3}{8}, \quad \langle \cos^2 \sin^2 \rangle = \frac{1}{8}, \\ \langle \cos^{2n} \rangle &= \langle \sin^{2n} \rangle = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)}, \quad n \geq 1. \end{aligned} \quad (54)$$

Other averages can either be derived from these, or found by direct integration. For instance,

$$\langle \cos^2 \sin^4 \rangle = \langle (1 - \sin^2) \sin^4 \rangle = \langle \sin^4 \rangle - \langle \sin^6 \rangle = \frac{3}{8} - \frac{15}{48} = \frac{1}{16}$$

and

$$\langle \cos^3 \sin \rangle = \frac{1}{2\pi} \int_0^{2\pi} \cos^3 \theta \sin \theta d\theta = -\frac{1}{2\pi} [\cos^4 \theta]_0^{2\pi} = 0.$$

EXAMPLE 7.6.3:

Consider the van der Pol equation $\ddot{x} + x + \varepsilon(x^2 - 1)\dot{x} = 0$, subject to the initial conditions $x(0) = 1$, $\dot{x}(0) = 0$. Find the averaged equations, and then solve them to obtain an approximate formula for $x(t, \varepsilon)$. Compare your result to a numerical solution of the full equation, for $\varepsilon = 0.1$.

Solution: The van der Pol equation has $h = (x^2 - 1)\dot{x} = (r^2 \cos^2 \theta - 1)(-r \sin \theta)$. Hence (53) becomes

$$\begin{aligned} r' &= \langle h \sin \theta \rangle = \langle (r^2 \cos^2 \theta - 1)(-r \sin \theta) \sin \theta \rangle \\ &= r \langle \sin^2 \theta \rangle - r^3 \langle \cos^2 \theta \sin^2 \theta \rangle \\ &= \frac{1}{2}r - \frac{1}{8}r^3 \end{aligned}$$

and

$$\begin{aligned} r\phi' &= \langle h \cos \theta \rangle = \langle (r^2 \cos^2 \theta - 1)(-r \sin \theta) \cos \theta \rangle \\ &= r \langle \sin \theta \cos \theta \rangle - r^3 \langle \cos^3 \theta \sin \theta \rangle \\ &= 0 - 0 = 0. \end{aligned}$$

These equations match those found in Example 7.6.2, as they should.

The initial conditions $x(0) = 1$ and $\dot{x}(0) = 0$ imply $r(0) \approx \sqrt{x(0)^2 + \dot{x}(0)^2} = 1$ and $\phi(0) \approx \tan^{-1}(\dot{x}(0)/x(0)) - \tau = 0 - 0 = 0$. Since $\phi' = 0$, we find $\phi(T) \equiv 0$. To find $r(T)$, we solve $r' = \frac{1}{2}r - \frac{1}{8}r^3$ subject to $r(0) = 1$. The differential equation separates to

$$\int \frac{8dr}{r(4-r^2)} = \int dT.$$

After integrating by partial fractions and using $r(0) = 1$, we find

$$r(T) = 2(1 + 3e^{-T})^{-1/2}. \quad (55)$$

Hence

$$\begin{aligned}
 x(t, \varepsilon) &\sim x_0(\tau, T) + O(\varepsilon) \\
 &= \frac{2}{\sqrt{1+3e^{-\varepsilon t}}} \cos t + O(\varepsilon).
 \end{aligned} \tag{56}$$

Equation (56) describes the transient dynamics of the oscillator as it spirals out to its limit cycle. Notice that $r(T) \rightarrow 2$ as $T \rightarrow \infty$, as in Example 7.6.2.

In Figure 7.6.4 we plot the “exact” solution of the van der Pol equation, obtained by numerical integration for $\varepsilon = 0.1$ and initial conditions $x(0) = 1$, $\dot{x}(0) = 0$. For comparison, the slowly-varying amplitude $r(T)$ predicted by (55) is also shown. The agreement is striking. Alternatively, we could have plotted the whole solution (56) instead of just its envelope; then the two curves would be virtually indistinguishable, like those in Figure 7.6.3. ■

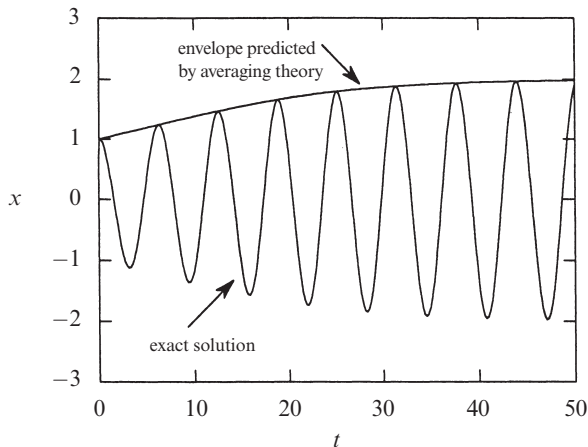


Figure 7.6.4

Now we consider an example in which the frequency of an oscillator depends on its amplitude. This is a common phenomenon, and one that is intrinsically *non-linear*—it cannot occur for linear oscillators.

EXAMPLE 7.6.4:

Find an approximate relation between the amplitude and frequency of the Duffing oscillator $\ddot{x} + x + \varepsilon x^3 = 0$, where ε can have either sign. Interpret the results physically.

Solution: Here $h = x^3 = r^3 \cos^3 \theta$. Equation (53) becomes

$$r' = \langle h \sin \theta \rangle = r^3 \langle \cos^3 \theta \sin \theta \rangle = 0$$

and

$$r\phi' = \langle h \cos \theta \rangle = r^3 \langle \cos^4 \theta \rangle = \frac{3}{8} r^3.$$

Hence $r(T) \equiv a$, for some constant a , and $\phi' = \frac{3}{8} a^2$. As in Example 7.6.2, the frequency ω is given by

$$\omega = 1 + \varepsilon \phi' = 1 + \frac{3}{8} \varepsilon a^2 + O(\varepsilon^2). \quad (57)$$

Now for the physical interpretation. The Duffing equation describes the undamped motion of a unit mass attached to a nonlinear spring with restoring force $F(x) = -x - \varepsilon x^3$. We can use our intuition about ordinary linear springs if we write $F(x) = -kx$, where the spring stiffness is now dependent on x :

$$k = k(x) = 1 + \varepsilon x^2.$$

Suppose $\varepsilon > 0$. Then the spring gets *stiffer* as the displacement x increases—this is called a **hardening spring**. On physical grounds we'd expect it to *increase* the frequency of the oscillations, consistent with (57). For $\varepsilon < 0$ we have a **softening spring**, exemplified by the pendulum (Exercise 7.6.15).

It also makes sense that $r' = 0$. The Duffing equation is a conservative system and for all ε sufficiently small, it has a *nonlinear center* at the origin (Exercise 6.5.13). Since all orbits close to the origin are periodic, there can be no long-term change in amplitude, consistent with $r' = 0$. ■

Validity of Two-Timing

We conclude with a few comments about the validity of the two-timing method. The rule of thumb is that the one-term approximation x_0 will be within $O(\varepsilon)$ of the true solution x for all times up to and including $t \sim O(1/\varepsilon)$, assuming that both x and x_0 start from the same initial condition. If x is a periodic solution, the situation is even better: x_0 remains within $O(\varepsilon)$ of x for *all* t .

But for precise statements and rigorous results about these matters, and for discussions of the subtleties that can occur, you should consult more advanced treatments, such as Guckenheimer and Holmes (1983) or Grimshaw (1990). Those authors use the *method of averaging*, an alternative approach that yields the same results as two-timing. See Exercise 7.6.25 for an introduction to this powerful technique.

Also, we have been very loose about the sense in which our formulas approximate the true solutions. The relevant notion is that of *asymptotic* approximation. For introductions to asymptotics, see Lin and Segel (1988) or Bender and Orszag (1978).

7.1 Examples

Sketch the phase portrait for each of the following systems. (As usual, r, θ denote polar coordinates.)

7.1.1 $\dot{r} = r^3 - 4r, \dot{\theta} = 1$

7.1.2 $\dot{r} = r(1 - r^2)(9 - r^2), \dot{\theta} = 1$

7.1.3 $\dot{r} = r(1 - r^2)(4 - r^2), \dot{\theta} = 2 - r^2$

7.1.4 $\dot{r} = r \sin r, \dot{\theta} = 1$

7.1.5 (From polar to Cartesian coordinates) Show that the system $\dot{r} = r(1 - r^2), \dot{\theta} = 1$ is equivalent to

$$\dot{x} = x - y - x(x^2 + y^2), \quad \dot{y} = x + y - y(x^2 + y^2),$$

where $x = r \cos \theta, y = r \sin \theta$. (Hint: $\dot{x} = \frac{d}{dt}(r \cos \theta) = \dot{r} \cos \theta - r \dot{\theta} \sin \theta$.)

7.1.6 (Circuit for van der Pol oscillator) Figure 1 shows the “tetrode multivibrator” circuit used in the earliest commercial radios and analyzed by van der Pol. In

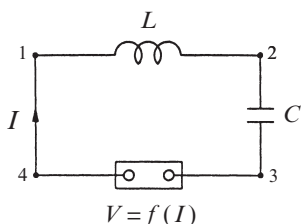


Figure 1

van der Pol's day, the active element was a vacuum tube; today it would be a semiconductor device. It acts like an ordinary resistor when I is high, but like a negative resistor (energy source) when I is low. Its current-voltage characteristic $V = f(I)$ resembles a cubic function, as discussed below.

Suppose a source of current is attached to the circuit and then withdrawn. What equations govern the subsequent evolution of the current and the various voltages?

- Let $V = V_{32} = -V_{23}$ denote the voltage drop from point 3 to point 2 in the circuit. Show that $\dot{V} = -I/C$ and $V = LI + f(I)$.
- Show that the equations in (a) are equivalent to

$$\frac{dw}{d\tau} = -x, \quad \frac{dx}{d\tau} = w - \mu F(x)$$

where $x = L^{1/2}I$, $w = C^{1/2}V$, $\tau = (LC)^{-1/2}t$, and $F(x) = f(L^{-1/2}x)$.

In Section 7.5, we'll see that this system for (w, x) is equivalent to the van der Pol equation, if $F(x) = \frac{1}{3}x^3 - x$. Thus the circuit produces self-sustained oscillations.

7.1.7 (Waveform) Consider the system $\dot{r} = r(4 - r^2), \dot{\theta} = 1$, and let $x(t) = r(t) \cos \theta(t)$. Given the initial condition $x(0) = 0.1, y(0) = 0$, sketch the approximate waveform of $x(t)$, *without* obtaining an explicit expression for it.